

The architecture of liquidity in the Italian electronic Market for Interbank Deposits

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Abstract

This study presents a longitudinal structural analysis of the Italian electronic Interbank Market (e-Mid) over a decade spanning the introduction of the Euro and the 2008 financial crisis. By integrating advanced graph theory visualization with Linear Mixed Model (LMM) econometrics, I document the existence of a hierarchical, scale-free network characterized by a "Robust-yet-Fragile" architecture. My analysis reveals a significant asymmetry in market integration: while domestic small banks are deeply embedded in the core, foreign institutions remain peripheral appendages. Econometric analysis using LMM reveals a dichotomy in pricing mechanisms: borrowing costs are driven by reputational factors (Eigenvector Centrality), whereas lending yields are determined by intermediation power (Betweenness Centrality). Furthermore, Principal Component Analysis (PCA) identifies a regime shift post-Euribor manipulation (started in 2005), where the structural profit advantages of core banks were eroded by a non-linear explosion in borrowing spreads, signalling a breakdown in the market's arbitrage mechanism during the financial crisis.

1. Introduction

The interbank market is the central nervous system of the modern financial infrastructure, facilitating liquidity redistribution and monetary policy transmission. However, the assumption that this market operates as a frictionless, homogenous plane has been definitively challenged by the financial crises of the early 21st century. Instead, these markets function as complex, adaptive networks where the specific position of a bank dictates its access to liquidity and its cost of funding.

This paper utilizes the extensive e-Mid dataset to map the topological evolution of the interbank network from January 1999 to December 2009. Unlike Over The Counter (OTC) markets, E-Mid Interbank Market offers a unique transparent view into the microstructure of lending and borrowing relationships. Specifically, I investigate how the "Core-Periphery" architecture creates a tiered system of access, how this structure creates asymmetric pricing power in borrowing versus lending, and how the network's structural integrity degraded during the transition from the stability of the early 2000s to the turmoil of the Euribor manipulation period (2005-2008), followed by the 2008 financial crisis.

2. Related Literature

The analysis of interbank market structure has undergone a profound evolution in recent decades, shifting from approaches based on the hypothesis of homogeneous and frictionless markets toward the application of Network Theory. Existing literature highlights that the topology of connections between financial institutions is not neutral; rather, it determines system stability and price formation.

The theoretical starting point for the analysis of interbank networks is the seminal work of [Allen & Gale \(2000\)](#), who demonstrated how network structure determines the system's resilience to liquidity shocks. The authors argue that a "complete" market structure (where every bank is connected to all others) maximizes risk sharing, whereas "incomplete" or fragmented structures can transform small liquidity shocks into systemic crises through contagion mechanisms. This structural vulnerability was further conceptualized by [Haldane \(2009\)](#), who introduced the definition of the global financial system as a "Robust-yet-Fragile" network. From this perspective, the concentration of connections and high

interconnectivity allow the system to absorb random idiosyncratic shocks but render it extremely fragile to shocks targeting central nodes (Hubs). [Boss et al. \(2004\)](#), analysing the Austrian banking system, provided one of the first empirical evidences of the Scale-Free nature of interbank networks, demonstrating that the degree distribution follows a Power Law. Specifically, regarding the Italian electronic market e-MID, [Iori et al. \(2008\)](#) conducted a fundamental topological analysis, highlighting that the market structure is heterogeneous and characterized by preferential attachment. Their work demonstrates that the e-MID network is not random but dominated by a few hubs that facilitate intermediation, confirming the results obtained in our visual analysis. Supporting the observed hierarchical structure, [Fricke & Lux \(2015\)](#) applied stochastic block models to e-MID data, formally identifying a Core-Periphery structure. Their results indicate the persistence of a stable "core" of banks (intermediaries) and a variable "periphery," suggesting that market tiering is an intrinsic structural feature rather than a transient one. A crucial stream of research for the present study links the topological position of banks to the cost of funding, moving beyond the assumption that interest rates depend exclusively on counterparty credit risk. Theoretically, [Rochet & Tirole \(1996\)](#) suggest that interbank relationships facilitate peer monitoring, creating reputational incentives. Empirically, [Temizsoy, Iori, and Montes-Rojas \(2015\)](#) tested this hypothesis specifically on e-MID. Their study demonstrates that centrality measures (particularly Eigenvector Centrality) have a significant and negative impact on interest rates: more central and better-connected banks succeed in obtaining funding at favourable rates. This result provides solid academic support for the findings of our econometric model, which identifies an inverse relationship between centrality and borrowing spreads. Similarly, in an international context, [Bech & Atalay \(2010\)](#) analysed the US Federal Funds market, confirming that network topology influences prices and that centrality acts as a determinant of market power. Finally, literature has documented how interbank market structure shifts drastically during periods of stress. [Gabrieli \(2011\)](#) analysed the microstructure of e-MID before and during the 2007-2008 financial crisis, highlighting how uncertainty regarding counterparty solvency leads to a contraction of maturities and network fragmentation. In parallel, [Brunetti et al. \(2019\)](#) showed how interconnectedness and correlation among banks tend to increase during crises, reducing diversification benefits and increasing the risk of systemic contagion. These contributions are essential for interpreting the structural regime shift and the explosion of funding costs highlighted by our longitudinal analysis (PCA) in the post-2005 period characterized by Euribor manipulation.

3. Methods, Materials, and Data

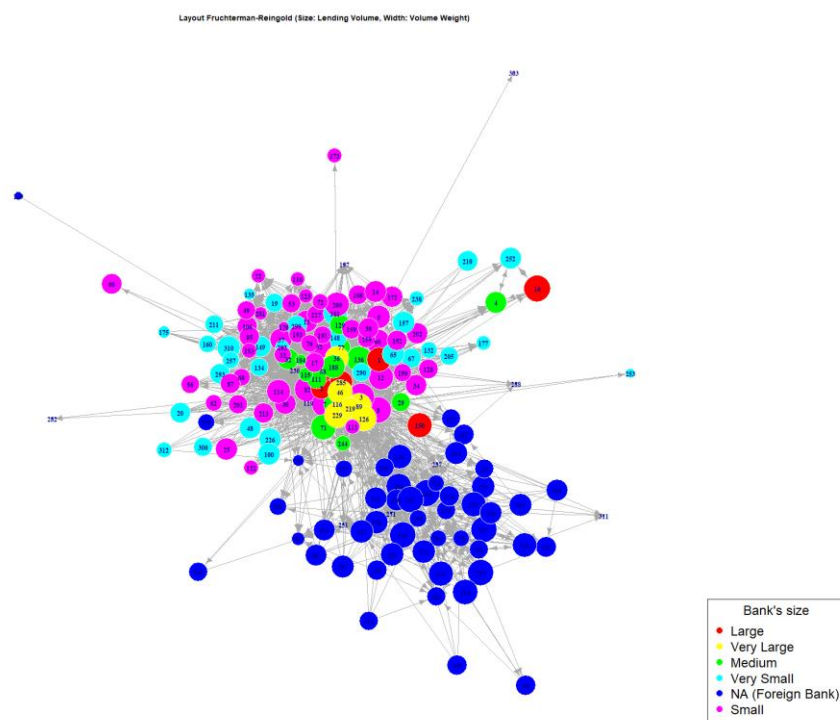
The study relies on monthly transaction data from Italian electronic Market for Interbank Deposits. For each bank I have its ID, the size, the number of lending and borrowing transactions, the traded volume, the lending and the borrowing spreads. Also, I have all the links between the banks (e.g. a link between bank 122 and bank 7 exist if a transaction occurred) and they are pointing from the lender to the borrower to follow the flow of liquidity. To handle the complexity of this dataset, I employ a multi-layered methodological approach: I utilize four distinct layouts (Fruchterman-Reingold, Kamada-Kawai, Circle, Sugiyama) to visualize the geometry of liquidity flows and assess spatial topology. Then, I display the degree distributions (K_{in} , K_{out}), node strength distribution (S_i) and the probability distribution of total degree with regards to a single snapshot of the network in April 2004 (pre-Euribor manipulation and before 2008 financial crisis). I analyse the probability distributions of clustering and diameter to quantify the network's heterogeneity along all the periods (1999-2009). I use Linear Mixed Models (LMM) to estimate borrowing and lending spreads, controlling for bank size groups, temporal trends, standardized centrality measures and unobserved bank heterogeneity via random effects. Finally, I employ Principal Component Analysis (PCA) to reduce the multidimensionality of network metrics into interpretable components (e.g. Network Structure vs. Systemic Importance), in the first instance, to

track bank profitability profiles at the single snapshot in April 2004 and in the second instance, to visualise temporal shifts in network structure from January 1999 to December 2009.

4. Results

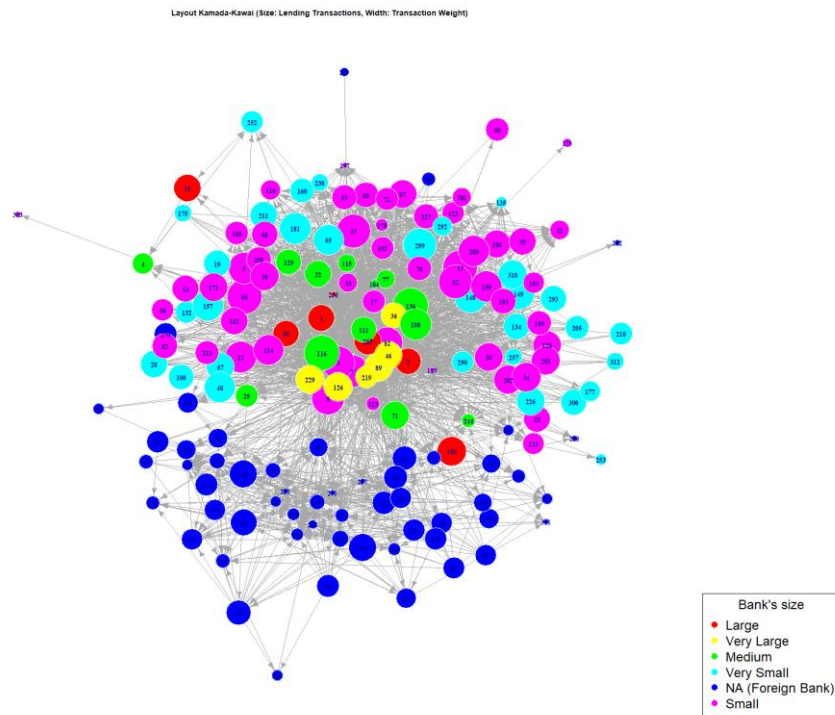
The application of distinct layout algorithms provides a composite view of the market's structural hierarchy. In Figure 1, the force-directed Fruchterman-Reingold layout simulates a physical system where connected nodes attract each other and unconnected nodes repel. This visualization immediately identifies a gravitational centre composed of "Very Large" (Yellow) and "Large" (Red) domestic banks. These entities possess the highest degree, pulling the system together. In stark contrast, a "Foreign Cloud" of blue nodes forms a massive, distinct cluster at the bottom right. Economically, this indicates that foreign banks are strictly peripheral; they do not trade broadly with the Italian system but hang off specific domestic hubs (gatekeepers). The domestic market appears highly interconnected, while the foreign segment is an appendage. This implies a vertical liquidity flow (center-to-periphery) rather than a horizontal peer-to-peer market.

Figure 1. Network structure in April 2008 (node size depends on lending volume, while edge width depends on the traded volume between the connected banks)



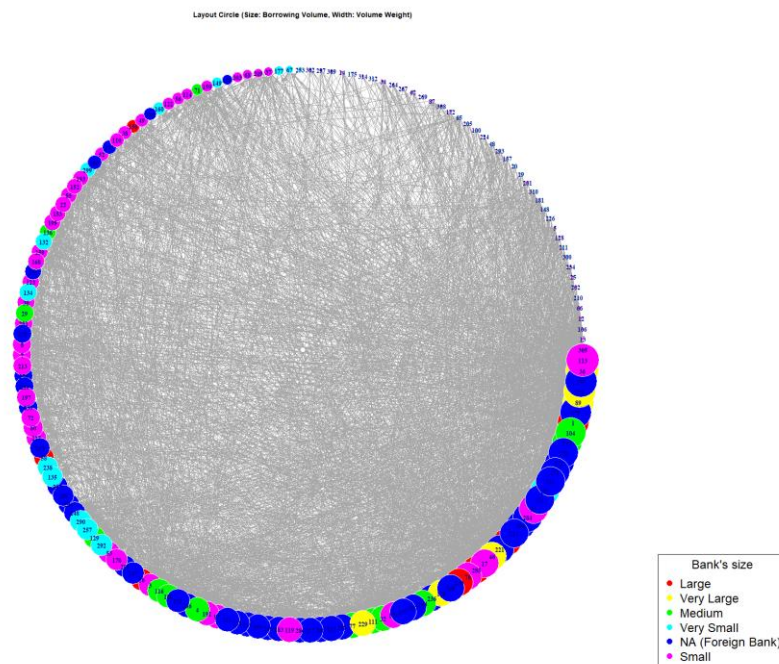
Complementing the core-periphery view, in Figure 2 the energy-based Kamada-Kawai algorithm attempts to position nodes so that the geometric distance between them is proportional to their graph-theoretic distance (shortest path). The graph appears as a compact, cohesive sphere, suggesting that structural distances are minimized and liquidity shocks can transmit rapidly across the system. Crucially, this layout challenges the notion that size equals centrality: many "Small" and "Very Small" domestic banks are embedded deep within the core. This suggests that small domestic banks are better integrated into the national liquidity system than their foreign counterparts, who remain marginalized. The nodes that sit in the middle of this layout are the "bridges", even if they aren't the biggest by size, they are crucial for connecting different clusters.

Figure 2. Network structure in April 2008 (node size depends on lending transactions, while edge width depends on the number of transactions between the connected banks)



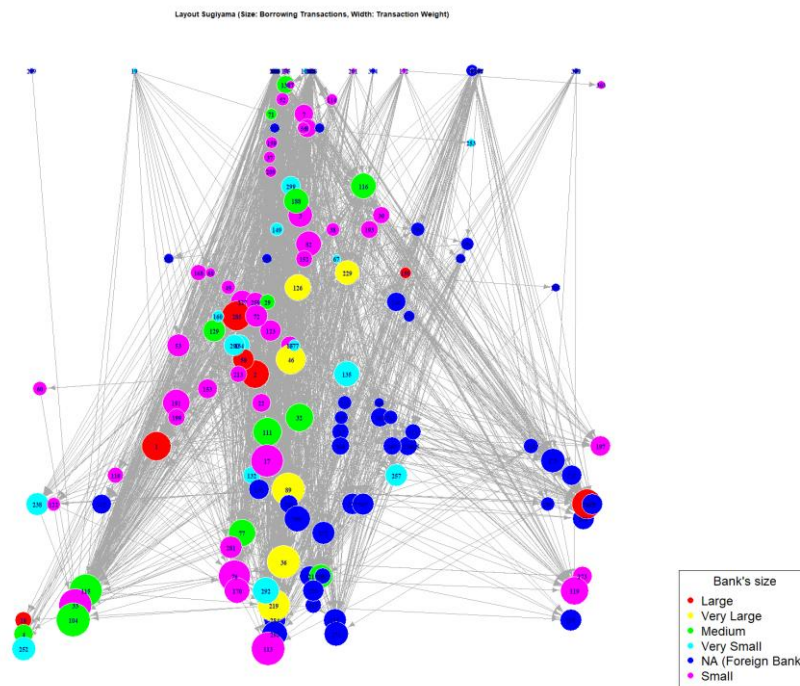
In Figure 3 the Circle layout highlights the sheer density of the market, a central "hairball" of grey lines indicating the market is highly liquid and active, and proving the system is a complex network, where a significant portion of the participants have direct links to many others (systemic complexity: a shock to one node has thousands of direct paths to propagate to others).

Figure 3. Network structure in April 2008 (node size depends on borrowing volume, while edge width depends on the traded volume between the connected banks)



In Figure 4 the Sugiyama hierarchical layout designed to show hierarchy and flow, usually placing sources at the top and sinks at the bottom. The plot shows a thick middle layer of intermediaries; this tiering system confirms that the market relies on a backbone of gatekeepers (high Betweenness) to distribute liquidity from pure lenders at the top to pure borrowers at the bottom.

Figure 4. Network structure in April 2008 (node size depends on borrowing transactions, while edge width depends on the number of transactions between the connected banks)



Statistical analysis of connectivity confirms that the E-Mid market is a scale-free system defined by extreme heterogeneity. Decomposing the connectivity reveals subtle asymmetries in how banks access and provide liquidity. Mathematically, the mean In-Degree (K_{in}) and Out-Degree (K_{out}) are identical ($\mu \sim 15.67$) because every loan granted is a loan received. However, the shapes of the distributions differ: In-Degree distribution (Figure 5) is sharply peaked at the lowest bin, resembling a Power Law. A vast number of banks (frequency > 60) have very few incoming connections ($K_{in} < 5$). This suggests that peripheral banks rely on a very small set of lenders for funding, highlighting a high dependency on specific counterparties. Out-Degree distribution (Figure 6) is also skewed and appears slightly "flatter" in the initial bins compared to the In-Degree. The tail, however, is significant and the comparison implies a tiered structure where a select group of banks (likely the "Large" and "Very Large" nodes identified in the network plots) act as money centre banks, providing liquidity to a wide array of borrowers.

Figure 5. In-Degree distribution of the network snapshot of April 2008

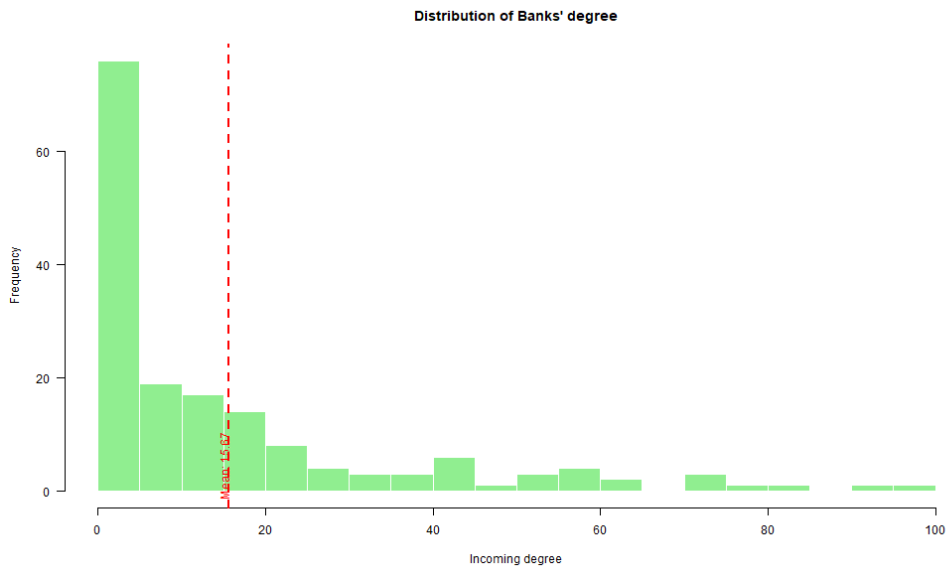
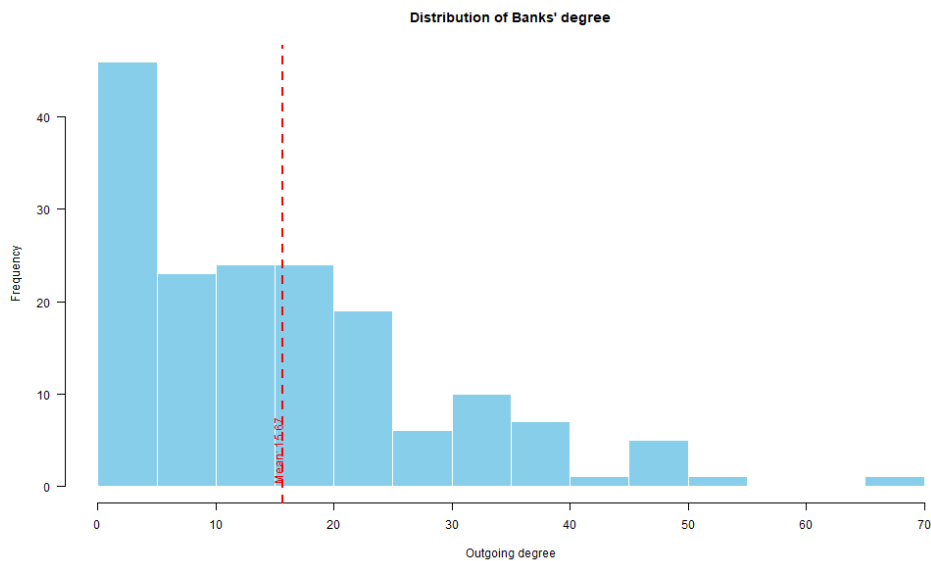


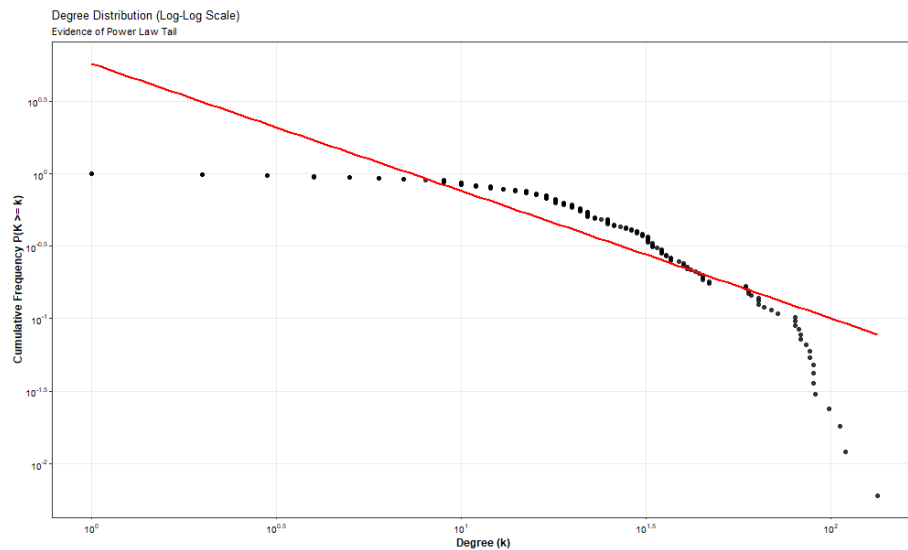
Figure 6. Out-Degree distribution of the network snapshot of April 2008



The structural organization of the interbank market is critically assessed through the analysis of the Total Degree distribution ($K_{in}+K_{out}$), defined as the sum of incoming and outgoing links for each node. The empirical data is visualized using a Complementary Cumulative Distribution Function (CCDF) plotted on a log-log scale (Figure 7), a standard method for detecting non-Gaussian topological features (A. Karpf et al., 2018). The distribution of degrees deviates significantly from a random graph model (Erdős-Rényi), which would exhibit a Poisson distribution peaked around the mean. Instead, the log-log plot reveals a highly skewed, heavy-tailed distribution. The linear decay observed in the tail of the log-log plot suggests that the probability of finding a node with degree k decays essentially as a power law distribution. The conformation of the empirical data (black dots) against the theoretical power-law fit (red line) in the CCDF plot elucidates the hierarchical nature of the market. The upper-left portion of the

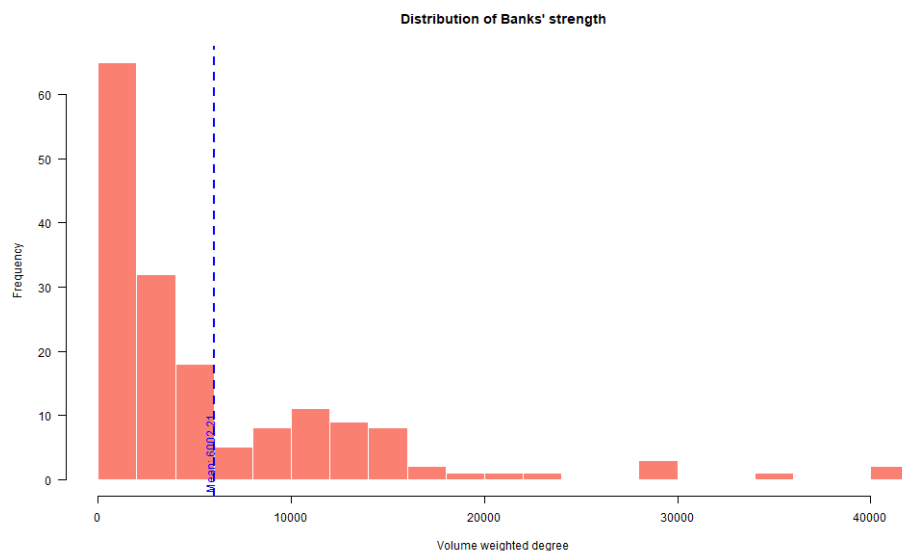
curve represents most banks. These institutions possess low connectivity and likely rely on a few specific counterparties for liquidity provision, corresponding to the "Periphery" identified in the topological layouts. The extension of the data points to the far right confirms the existence of "Hubs": these nodes occupy the extreme tail of the distribution. In network theory, these hubs act as the structural "glue" of the system, facilitating the short average path lengths (Small World property) observed in the subsequent diameter analysis.

Figure 7. Total Degree distribution of the network snapshot of April 2008



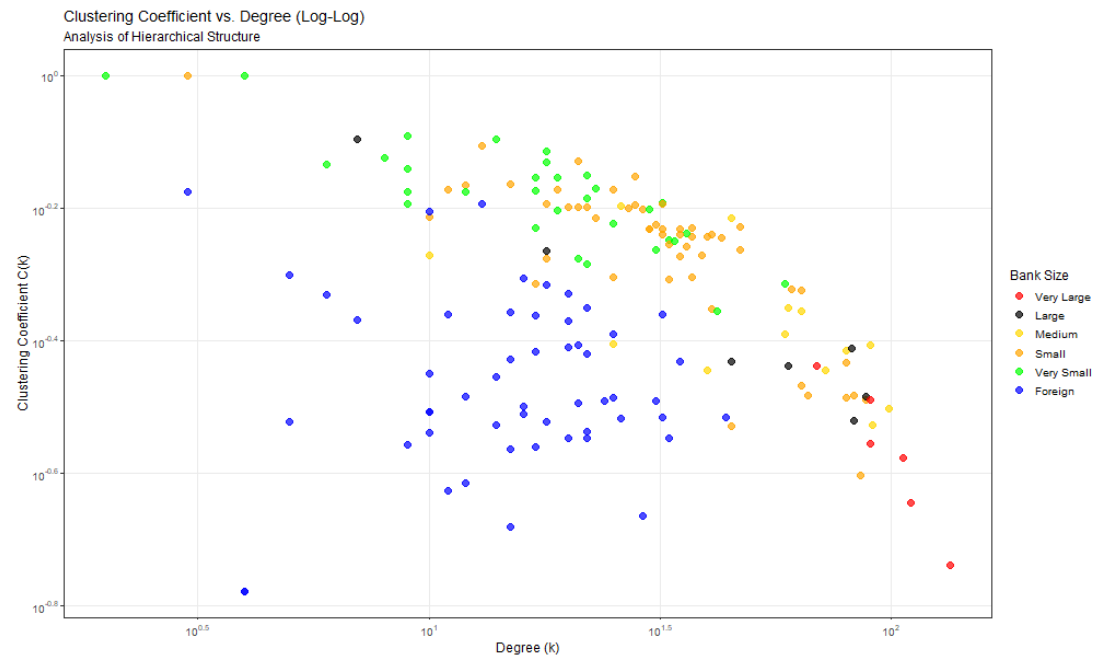
While Degree measures the number of partners, Strength (S_i) measures the total volume of transactions handled by a bank. The Strength distribution (Figure 8) displays the most extreme skewness among all metrics. The distribution approximates a Scale-Free property more aggressively than the topology. The discrepancy between most small banks and the few "hubs" suggests that liquidity is far more concentrated than connectivity: a bank might have many connections (high degree), but the actual volume of money flows through a very small subset of these links.

Figure 8. Strength distribution of the network snapshot in April 2008



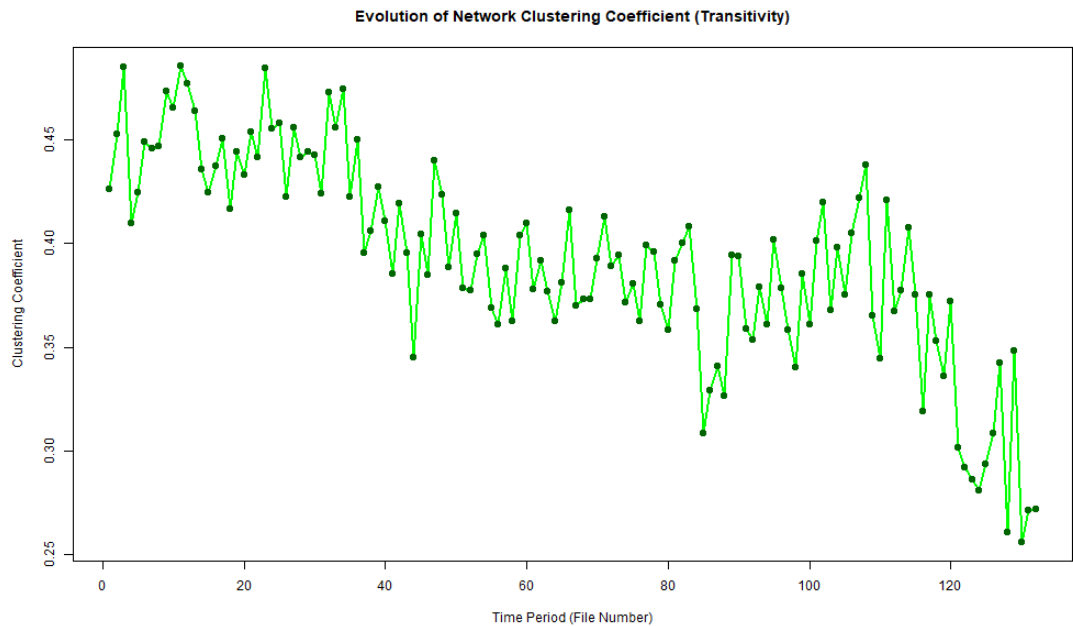
In Figure 9, the analysis of the Clustering Coefficient against Degree ($C(k)$) provides the final proof of a hierarchical architecture (Karpf et al., 2018): I observe a negative correlation where low-degree nodes form dense communities (high clustering), while high-degree hubs exhibit low clustering (star-like topology). This confirms that "Very Large" banks act as global bridges connecting disparate local communities of small banks.

Figure 9. Clustering Coefficient distribution against Degree (April 2008)



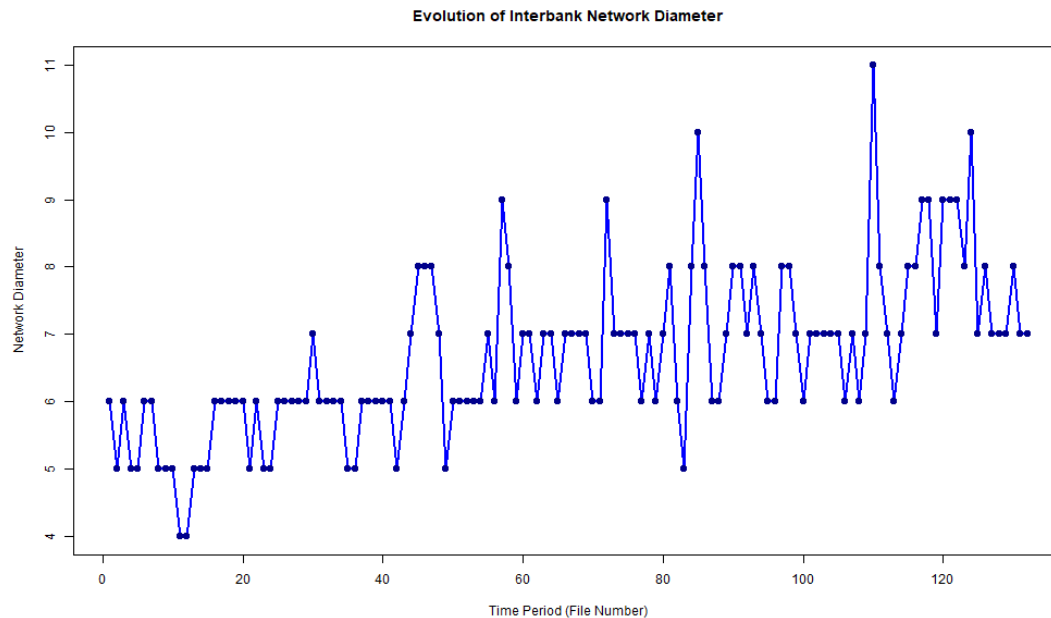
Longitudinal analysis over all the 132 months shows a progressive deterioration in the network's structural health. In Figure 10, a high clustering coefficient often correlates with network resilience, as triangular relationships provide alternative pathways for liquidity. The observed decline signals a potential increase in structural fragility; as the network becomes less "clustered," the reliance on central hubs likely increases, and the failure of a specific link becomes harder to circumvent locally.

Figure 10. Evolution of network Clustering Coefficient (1999-2009)



The increase in diameter (Figure 11) represents a frictional cost to the system. In a financial context, a larger diameter implies that shocks in one part of the network take longer to propagate by the rest of the system. It suggests a segmentation of the market where peripheral banks are becoming topologically more distant from the core liquidity providers.

Figure 11. Evolution of network Diameter (1999-2009)



Linear Mixed Models (Table 1 and Table 2) on the longitudinal dataset spanning the full 132 months period uses as dependent variable the borrowing spread or the lending spread. The fixed effects include the standardized centrality metrics (In-Degree, Eigenvector, Betweenness) calculated over the entire period, and a categorical variable 'Group' capturing bank size. Crucially, the model incorporates a time trend to control for macroeconomic fluctuations and random intercepts at the bank level to account for unobserved idiosyncratic heterogeneity and serial correlation.

Table 1. Regression analysis using borrowing spread as dependent variable (1999-2009)

```
Formula: borrowing_spread ~ Group_Factor + Month_Centered + In_Degree_Std +
Eigen_Std + Betw_Std + (1 | ID)
Data: df_borr

      AIC      BIC   logLik deviance df.resid
-53072.0 -52976.2  26548.0 -53096.0    21645

Scaled residuals:
      Min       1Q   Median       3Q      Max
-15.4837  -0.2608  -0.0213   0.1738   21.1215

Random effects:
 Groups Name      Variance Std.Dev.
 ID      (Intercept) 0.0005335 0.02310
 Residual              0.0049012 0.07001
Number of obs: 21657, groups: ID, 351

Fixed effects:
              Estimate Std. Error      df t value Pr(>|t|)
(Intercept)    3.241e-02  4.546e-03  1.439e+03   7.130 1.58e-12 ***
Group_FactorForeign -4.078e-02  5.349e-03  9.143e+02  -7.624 6.15e-14 ***
Group_FactorMedium -2.386e-02  4.613e-03  3.858e+03  -5.172 2.44e-07 ***
Group_FactorSmall -2.354e-02  4.783e-03  1.949e+03  -4.921 9.35e-07 ***
Group_FactorVery Large -1.656e-02  8.168e-03  9.986e+02  -2.027  0.0429 *
Group_FactorVery Small -2.498e-02  4.827e-03  2.085e+03  -5.175 2.49e-07 ***
Month_Centered    1.062e-02  6.063e-04  1.576e+04  17.513 < 2e-16 ***
In_Degree_Std     1.191e-02  8.046e-04  1.393e+04  14.807 < 2e-16 ***
Eigen_Std        -4.081e-03  5.860e-04  2.103e+04  -6.965 3.39e-12 ***
Betw_Std          6.081e-04  7.223e-04  2.096e+04   0.842  0.3999
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The coefficient for `In_Degree_Std` is positive and highly significant: contrary to the intuition that "more connections are better," increasing the number of lenders is associated with a higher borrowing spread. This suggests a "desperation effect" or high search costs: banks that are forced to borrow from a multitude of counterparties may be perceived as financially fragile or unable to secure sufficient liquidity from prime lenders, thus paying a premium. `Eigen_Std` exhibits a negative and significant coefficient: banks connected to other highly central nodes benefit from a reduced cost of funding. It implies that being part of the "core club" of interconnected banks signals trustworthiness, allowing these institutions to borrow at discounted rates compared to the market average.

Table 2. Regression analysis using lending spread as dependent variable (1999-2009)

```
Formula: lending_spread ~ Group_Factor + Month_Centered + In_Degree_Std +
Eigen_Std + Betw_Std + (1 | ID)
Data: df_lend

      AIC      BIC    logLik deviance df.resid
-41066.5 -40970.7  20545.3 -41090.5    21645

Scaled residuals:
      Min       1Q   Median       3Q      Max
-12.2146  -0.2637  -0.0562   0.1153   20.2151

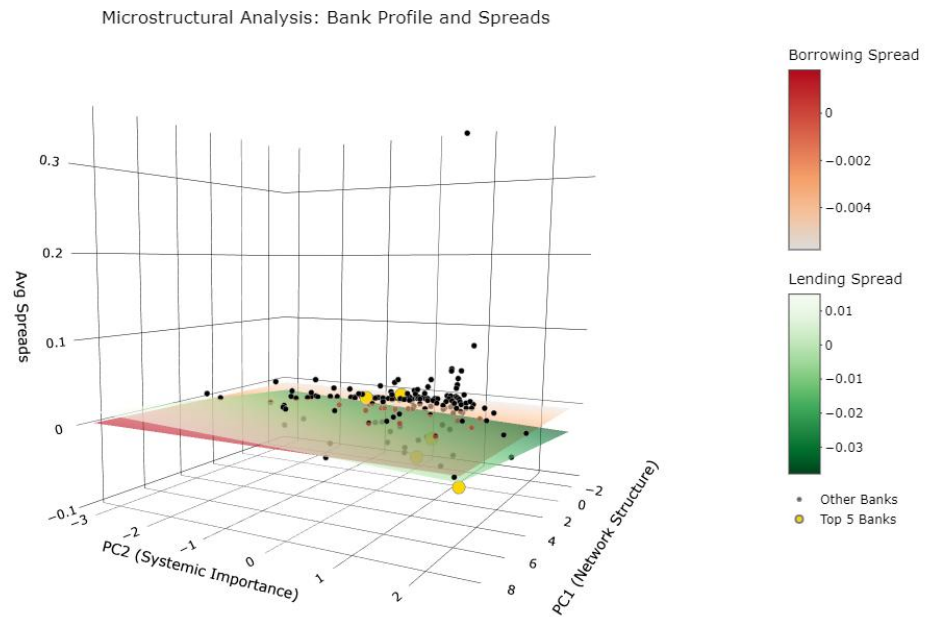
Random effects:
 Groups   Name                Variance Std.Dev.
 ID       (Intercept)         0.001466 0.03829
 Residual                            0.008482 0.09210
Number of obs: 21657, groups: ID, 351

Fixed effects:
              Estimate Std. Error    df t value Pr(>|t|)
(Intercept)    1.389e-02  6.520e-03 1.771e+03  2.130  0.03334 *
Group_FactorForeign -1.553e-02  7.889e-03 9.378e+02 -1.968  0.04937 *
Group_FactorMedium  1.978e-02  6.359e-03 6.192e+03  3.110  0.00188 **
Group_FactorSmall   5.290e-03  6.747e-03 3.004e+03  0.784  0.43304
Group_FactorVery Large -9.017e-03  1.190e-02 1.372e+03 -0.758  0.44870
Group_FactorVery Small  2.133e-02  6.794e-03 3.148e+03  3.140  0.00171 **
Month_Centered   1.251e-02  8.098e-04 1.831e+04 15.443 < 2e-16 ***
In_Degree_Std    -2.204e-03  1.078e-03 1.658e+04 -2.044  0.04093 *
Eigen_Std        -2.637e-04  7.756e-04 2.144e+04 -0.340  0.73384
Betw_Std         4.163e-03  9.560e-04 2.151e+04  4.355 1.34e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Analysis of Variance (ANOVA) on both borrowing and lending side provides compelling statistical evidence favouring the inclusion of network centrality measures. The lending side of the market operates on a different logic. Here, Eigenvector centrality is irrelevant. Instead, Betweenness Centrality becomes a positive and significant driver. This suggests that banks acting as bridges exercise market power, charging a markup to borrowers who cannot access the core directly. Furthermore, high Degree leads to lower yields in lending, indicating that large market makers compete aggressively on volume rather than margin.

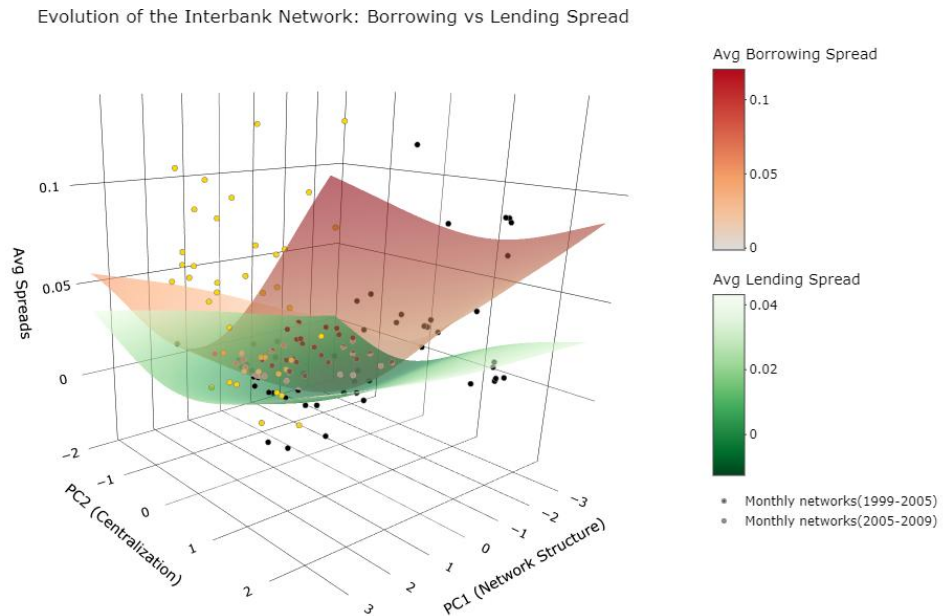
In Figure 12, Principal Component Analysis (PCA) projects the multidimensional network metrics into a 2D space defined by PC1 and PC2. In the stable period (April 2008), "Top Performers" (banks maximizing the difference between lending and borrowing spread) are clustered in the high-centrality region of the PCA plot suggesting that these banks exploited an "efficiency frontier," leveraging their reputation to borrow cheap and their market power to lend dear.

Figure 12. PCA plot of the network snapshot of April 2008



On the other hand, in Figure 13 the longitudinal analysis reveals a regime shift post-2005 (yellow points). The response surfaces show a non-linear explosion in the borrowing spread that deformed the market structure. While borrowing costs skyrocketed due to liquidity hoarding, lending yields remained relatively inelastic. This asymmetric shock compressed the intermediation margin, effectively breaking the arbitrage mechanism that had sustained the "Top Performers" in the pre-crisis era. The network topology, once a source of profit, became a transmitter of stress.

Figure 13. PCA plot of the network evolution over 132 months



5. Conclusions

The investigation of the E-Mid market from 1999 to 2009 leads to three critical conclusions regarding the relationship between network structure and financial stability. First, the market is structurally "Robust-yet-Fragile": the scale-free degree distribution and the dependence on a few "Very Large" domestic hubs created a system resilient to random peripheral failures but extremely vulnerable to the crisis of core nodes. The marginalization of foreign banks into a distinct "cloud" further exacerbated this by creating a dependent appendage rather than an integrated European market. Second, pricing power is structurally determined and asymmetric. Borrowing is a reputation game where "who you know" (Eigenvector) reduces costs, while lending is a power game where "where you sit" (Betweenness) increases yields. This confirms that the interbank market is a hierarchy where topology dictates terms. Finally, the network is highly robust against the random exit of banks. Since most nodes are peripheral (low degree), a random shock is statistically likely to hit a non-systemic bank, causing negligible damage to the overall connectivity. Conversely, the network is extremely fragile to shocks targeting the Hubs (the tail nodes). This confirms that the "Too Big to Fail" problem is a structural feature of the network topology itself.

References

- Allen, F., & Gale, D. (2000). Financial Contagion. *Journal of Political Economy*, 108(1). pp. 1—33.
- Haldane, A. G. (2009). Rethinking the financial network. *Speech delivered at the Financial Student Association, Amsterdam*. pp. 1–25
- Boss, M., Elsinger, H., Summer, M., & Thurner, S. (2004). Network topology of the interbank market. *Quantitative Finance*, 4(6). pp. 677—684
- Iori, G., De Masi, G., Previati, A., Gabbi, G., & Caldarelli, G. (2008). A network analysis of the Italian overnight money market. *Journal of Economic Dynamics and Control*, 32(1). pp. 265—270.
- Fricke, D., & Lux, T. (2015). Core–periphery structure in the overnight money market: evidence from the e-MID trading platform. *Computational Economics*, 45(3). pp. 365—372
- Rochet, J. C., & Tirole, J. (1996). Interbank lending and systemic risk. *Journal of Money, Credit and Banking*, 28(4). pp. 734—740
- Temizsoy, A., Iori, G., & Montes-Rojas, G. (2015). Network centrality and funding rates in the e-MID interbank market. *Journal of Financial Stability*, 19. pp. 1—12
- Bech, M. L., & Atalay, E. (2010). The topology of the federal funds market. *Physica A: Statistical Mechanics and its Applications*, 389(22). pp. 5244—5245
- Gabrieli, S. (2011). The microstructure of the money market before and during the financial crisis: a network analysis. *CEIS Research Paper No. 181*. pp.20—25
- Brunetti, C., Harris, J. H., Mankad, S., & Michailidis, G. (2019). Interconnectedness in the interbank market. *Journal of Financial Economics*, 133(2). pp. 522-530
- Karpf, A., Mandel, A., Battiston, S. *Journal of Economic Behavior and Organization* 153. pp. 103—122

